Effectiveness of alcohol-based hand rubs for removal of Clostridium difficile spores from hands.

BACKGROUND: Alcohol-based hand rubs (ABHRs) are an effective means of decreasing the transmission of bacterial pathogens. Alcohol is not effective against Clostridium difficile spores. We examined the retention of C. difficile spores on the hands of volunteers after ABHR use and the subsequent transfer of these spores through physical contact. METHODS: Nontoxigenic C. difficile spores were spread on the bare palms of 10 volunteers. Use of 3 ABHRs and chlorhexidine soap-and-water washing were compared with plain water rubbing alone for removal of C. difficile spores. Palmar cultures were performed before and after hand decontamination by means of a plate stamping method. Transferability of C. difficile after application of ABHR was tested by having each volunteer shake hands with an uninoculated volunteer. RESULTS: Plain water rubbing reduced palmar culture counts by a mean (+/- standard deviation [SD]) of 1.57 +/- 0.11 log10 colony-forming units (CFU) per cm2, and this value was set as the zero point for the other products. Compared with water washing, chlorhexidine soap washing reduced spore counts by a mean (+/- SD) of 0.89 +/- 0.34 log10 CFU per cm2; among the ABHRs, Isagel accounted for a reduction of 0.11 +/- 0.20 log10 CFU per cm2 (P = .005), Endure for a reduction of 0.37 +/- 0.42 log10 CFU per cm2 (P = .010), and Purell for a reduction of 0.14 +/- 0.33 log10 CFU per cm2 (P = .005). There were no statistically significant differences between the reductions achieved by the ABHRs; only Endure had a reduction statistically different from that for water control rubbing (P = .040). After ABHR use, handshaking transferred a mean of 30% of the residual C. difficile spores to the hands of recipients. CONCLUSIONS: Hand washing with soap and water is significantly more effective at removing C. difficile spores from the hands of volunteers than are ABHRs. Residual spores are readily transferred by a handshake after use of ABHR.